



Adam Arras

Mathematics | Probability & Stochastic Processes | Deep Learning

Researcher in mathematics with expertise in stochastic processes and random graphs, combined with hands-on experience in Python-based deep learning and scientific consulting for industrial R&D projects.

Experience

- 2025–Present **Scientific Consultant**, *Research To Business*, Paris
- Analysis of industrial R&D projects in AI, software and data science.
 - Preparation of technical R&D reports, project summaries and scientific syntheses for innovation funding applications.
- 2023–Present **Postdoctoral Researcher**, *Alfréd Rényi Institute of Mathematics*, Budapest
- Research on Markov chains, spectral theory, and graph limits arising from large-scale networks.
 - Supported by the ERC Synergy project Dynamics and Structure of Networks (DYNASNET).
- 2022–2023 **Teaching and Research Assistant (ATER)**, *Aix-Marseille University*, Marseille
- Conducted mathematical research in spectral theory of large random graphs.
 - Taught mathematics courses to engineering students at Polytech Marseille; designed course materials, problem sets and assessments.

Technical Skills

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| Programming | Python: NumPy, Pandas, Matplotlib, NetworkX, PyTorch |
| Development | GitHub, Jupyter Notebook, Google Colab |
| Scientific writing | Quarto, Markdown, L ^A T _E X |

Projects

- 2026 **Deep Learning mini-course**: Python/Jupyter notebooks on deep learning, including architectures implemented from scratch and training diagnostics.
- Course page: adamarras.github.io/posts/posts-2
 - Materials: github.com/AdamArras/mini-course-DL

Education

- 2019–2023 **PhD in Mathematics**, *Institut de Mathématiques de Marseille*, Aix-Marseille University. Supervised by Charles Bordenave. Topics: probability, random graphs and spectral theory.

- 2015–2019 **Normalien**, *ENS Paris-Saclay* (formerly ENS Cachan), France
- Admitted through a highly competitive national entrance examination.
 - Research internships in applied mathematics: image processing at CMLA, ENS Paris-Saclay; Monte Carlo methods at the Department of Statistics, University of Oxford; and community detection at LANEAS, CentraleSupélec.
- 2018–2019 Predoctoral research visit, Georgia Institute of Technology, Atlanta, USA.
- 2017–2018 MSc in Mathematics, Computer Vision and Machine Learning, Master MVA, ENS Paris-Saclay.

Selected Publications

- M. Abért, A. Arras and J. Kim, *Quenched Path Limits and Periodization Stability for Tilted Brownian Motion in Poissonian Potentials on $\mathbb{H}^d$* , arXiv:2602.18433 (2026).
- M. Abért, A. Arras and J. Kim, *The Uniform Random Walk on Graphs, Loop Processes and Graphings*, arXiv:2512.09166 (2025).
- A. Arras and C. Bordenave, *Existence of absolutely continuous spectrum for Galton–Watson random trees*, *Communications in Mathematical Physics* 403 (2023), no. 1, 495–527.

Selected Talks

- 2026 GDR Matrices et Graphes Aléatoires (MEGA); Probability Seminar, MODAL’X; Research School LOUCCOUM, CIRM.
- 2023–2024 Research Seminar, Alfréd Rényi Institute of Mathematics; CMI-HIMR Summer School on Symmetry and Randomness, University of Bristol; Probability Seminar, IRMA; Workshop in Discrete Mathematics and Probability, Alfréd Rényi Institute of Mathematics.
- 2021–2022 Probability Seminar, UMPA; LdRAM Working Group, ENS Lyon; GDR MEGA Probability Summer Schools, Aussois and Saint-Malo; Colloque Jeunes Probabilistes et Statisticiens, Oléron.

Scientific Activities

- 2019–2026 Graduate-level teaching in mathematics for engineering and university students.
Co-organizer of scientific workshops and academic events.
Peer reviewer for academic journals *Electronic Communications in Probability* and *European Journal of Combinatorics*.
- 2022 Scientific outreach for the project Regards de Géomètre, Les maths en scènes.

Languages

French	Native
English	Proficient
German	Basic